

ENG 101

Lesson 18

Primary and Secondary memory[1] The term 'memory' is usually used to refer to the **internal storage** locations of a computer. It is also called **real storage** or primary memory, and is expressed as quantities of K. For example, computers are advertised as having memories of 16K or 152k, depending on their storage capacity. Each k is equal to 1,024 **bytes**, and each byte is equal to 8 **bits**.

[2] Primary memory is closely associated with the CPU because it stores programs and data temporarily, thus making them immediately available for processing by the CPU. To facilitate processing, two things are needed: **random access** and speed. The former means that any part of the memory may be read, or **accessed**, equally quickly. This is made possible by the system of **addresses** in primary memory, where the storage locations are like a series of tiny compartments, each having its own address.

These addresses are like the addresses of houses, in that they do not change. Because they are always fixed, the control unit knows where to find them at a very high speed. When it finds them, it puts into the compartments whatever must go there and wipes out whatever was stored there. The information present in these compartments is called the **contents** of the memory.

[3] Most primary memory is costly, and therefore it is used transiently, which means that a program, or parts of it, is kept in internal storage while the program is being executed. This, however, is not true for mini and micro applications where the computer performs the same function, referred to as a **dedicated function**, all the time. But since computers must process vast quantities of data and programs, a lot of storage space is required. For this reason various secondary memory technologies have been developed.

[4] Secondary memory devices fall into two categories: **sequential devices** and **random-access** devices. Sequential devices permit information to be written onto or read off some storage medium in a fixed sequence only. In order to get at a particular data item, it is necessary to pass over all the data preceding it. An example of such a device is the **magnetic tape**. It's cost is low, but access to specified data may take a considerable length of time.

On the other hand, random-access devices are designed to permit direct, or almost direct, access to specified data. These devices bypass large quantities of irrelevant data and therefore reduce access time considerably. An example of this technology is the magnetic disk, which is faster than the magnetic tape and also more expensive.

When disks are hooked up to the computer and used as an extension of internal storage in order to increase the capacity of primary memory, this is called virtual storage. For example, a computer with 256K bytes of real storage may seem to have 512K bytes of virtual storage by using disks to provide additional storage.

1 Main idea :

Which statement best expresses the main idea of the text? Why did you eliminate the other choices?

1. There are two types of memory: primary and secondary.
2. Primary memory is more important than secondary memory.
3. Secondary memory devices are unimportant in a computer system.

2 Understanding the passage:

Indicate whether the following ideas were stated or not stated (S/NS) in the text.

S NS

☐ ☐

1. The term 'memory' can be expressed in other ways.

- ☐ ☐ 2. Computers are often advertised according to their memory capacity.
- ☐ ☐ 3. The CPU can easily access information from internal storage.
- ☐ ☐ 4. Minicomputers and microcomputers have a similar memory capacity.
- ☐ ☐ 5. The control unit needs to know the location where information is stored or needs to be stored.
- ☐ ☐ 6. Primary memory is more expensive than secondary memory.
- ☐ ☐ 7. There are two types of secondary memory device.
- ☐ ☐ 8. Information stored on magnetic disk can be retrieved faster than if that same information were on tape.
- ☐ ☐ 9. Disks and tapes can be stored in a library.
- ☐ ☐ 10. Computers can process information even if complete programs are not put in internal storage.

3 Locating Information

Find the passages in the text where the following ideas are expressed. Give the paragraph reference.

-1. Speed and random access are important in processing information.
-2. Random-access devices are more efficient than sequential devices.
-3. The CPU and primary memory work closely together.
-4. Virtual storage increases the memory capacity of a computer.
-5. Real storage, internal storage, and primary memory are all the same
-6. Information is stored in memory in compartments with a specific location.
-7. There are two classes of secondary memory device.
-8. Only parts of programs are kept in primary storage while a program is being run through.

4 Contextual reference

Look back at the text and find out what the words in **bold** typeface refer to.

- 1. **It** is also called
- 2. depending on **their** storage capacity
- 3. Thus making them
- 4. **The former** means that
- 5. **where** the storage locations
- 6. **each** having its own address
- 7. in that **they** do not change
- 8. where to find **them**
- 9. whatever must go **there**
- 10. or parts of **it**

5 Understanding words

Refer back to the text and find synonyms for the following words.

- 1. represented
- 2. erases

3. carried out
4. before
5. very much

Now refer back to the text and find antonyms for the following words.

6. latter
7. disallow
8. unnecessary
9. go through
10. imaginary

6 Word forms

First choose the appropriate form of the words to complete the sentences. Then check the differences of meaning in your dictionary.

1. expression, expressive, express, expressed
 - a. Information sent via a computer is faster than using the.....system of airlines or postal services.
 - b. An.....such as 'He or she has a computer of a brain' means that he or she is a fast-thinking person.
 - c. Computers understand commands.....in the form of 0 and 1.
2. equality, equal, equally, equalize
 - a. The symbol # means that two things are not..... . b. A microcomputer doesn't.....a minicomputer in flexibility.
 - c. The two computer languages Pascal and PL1 are.....difficult.
3. consideration, consider, considerable, considerably
 - a. There is a.....difference between written and spoken English.
 - b. It is important to.....the capabilities and limitations of a computer before buying one.
 - c. New printers can print results.....faster than previously.
4. design, designed, designation, designer, designing
 - a. Due to the advances in computer technology, computer.....are faced with a more challenging job.
 - b. Computers are.....to process information accurately and quickly.
 - c. Computer architects are constantly trying to improve on the.....of computers.
5. advertisement, advertise, advertised
 - a. There are many computer-related jobs.....in The New York Times.
 - b. The Computers Center will soon.....for more operators and programmers.
 - c. Career opportunities in computer science and related fields can usually be found in the.....section of newspapers.

7 Content review

Match the words in column A with the words or statements in column B.

A	B
☐ 1. internal storage quickly	a. means any part of memory can be read equally
☐ 2. real storage	b. the information contained in the storage locations
☐ 3. random access	c. are storage locations in internal storage
☐ 4. addresses locations	d. refers to memory contained in the storage

- ☐ 5. Contents e. hooking up secondary memory devices onto memory to increase their capacity
- ☐ 6. sequential access f. sometimes called primary memory
- ☐ 7. Virtual storage g. information must be read from secondary memory devices in a fixed pattern

8 Focus review

We learnt in earlier lesson how to use examples to explain a point or to illustrate an idea in a given text. It is important to learn to differentiate between the idea and the illustration of the idea with examples. Writers often say explicitly which things are examples by using connectives like *for example*, *such as*, *etc...*

Complete the following table by referring back to the text on 'Primary and Secondary Memory', and locating connectives used for illustrations and examples.

Para	Items(s) to be Exemplified	Example Marker	Example
[1]	quantities of K		
[4]a.		an example	
[4]b.			Magnetic disk
[4]c.	virtual storage		

Reinforcement (e.g)

1. **In addition** to their speed ,computers are accurate and can do repetitive operations over and over again without becoming tired or bored .

2. Microcomputers are cheaper than mainframes , **as well as** being compact and portable .

Similarity (e.g)

1. Microcomputers can have a storage capacity of up to 32k, **likewise** minicomputers .

2. Many minicomputers are used merely for a fixed application and run only a single program . Microcomputers operate **in much the same way** as is the case in automobile emission control systems.